

Roll Number: _____

Thapar Institute of Engineering & Technology
Department of Electronics and Communication Engineering

M.E. (II SEM): VLSI Design

Course Code: PVL333

Course Name: System On Chip

May 25, 2024, 2:00 PM

Time: 03 Hrs.

Max. Marks: 50.

Name of Faculty: Dr. Robin Singla

Note: Attempt any 5 questions & assume missing data, if any, suitably

Q. No.	Questions	Marks	CO	BL
Q.1	Discuss different integration platforms in SOC design.	(10)	C05	L2 & L3
Q.2	Explain different types of system level power estimation and modeling approaches.	(10)	C03	L2 & L3
Q.3	With the help of a block diagram explain the transaction level modeling.	(10)	C04	L2 & L3
Q.4	Explain SOC design methodologies.	(10)	C05	L2 & L3
Q.5	Explain in detail the software design in SOC. What are the different issues associated with software design in SOC.	(10)	C03	L2 & L3
Q.6	Explain SOC test scheduling and test integration.	(10)	C05	L2 & L3

Q.1	Discuss different integration platforms in SOC design.	(10)
	During the SoC integration process, SoC Compiler automates the management of power intent through an automated generation, update, promotion and demotion of UPF files	
Q.2	Explain different types of system level power estimation and modeling approaches.	(10)
	The power consumption models can be distinguished into two main categories: - Low-Level or hardware level models. ➤ Circuit/Transistor level ➤ Logic gate level ➤ RTL ➤ Architectural level High-Level models. ➤ Instruction Level Power Analysis (ILPA) ➤ Functional Level Power Analysis (FLPA).	
Q.3	With the help of a block diagram explain the transaction level modeling.	(10)
	<pre> graph TD CS[Customer Specification] --> PS[Paper Specification] PS --> HSWP[HW/SW Partitioning] HSWP --> TLM[TLM] TLM --> HD[Hardware Development] TLM --> SD[Software Development] HD --> CHWSE[Concurrent HW/SW Engineering Based on TLM] SD --> CHWSE CHWSE --> TC([Test Chip]) TC --> SIV([System Integration & Validation]) SIV --> CF[Chip Fabrication] </pre>	
Q.4	Explain SOC design methodologies.	(10)
	• The primary design methods used today can be divided into three segments: ▪ Timing Driven Design (TDD) ▪ Block Based Design (BBD) ▪ Platform Based Design (PBD)	

	• These segments vary depending on the ▪ Linchpin technologies used, ▪ Design capacity, ▪ Level of and investment in design reuse.	
Q.5	Explain in detail the software design in SOC. What are the different issues associated with software design in SOC.	(10)
	• Critical components of a software development environment: ▪ Compiler for efficient code generation, predictable execution times of tasks and processes e.g. gcc, g++, javac. ▪ Interfaces for generation of device drivers for external units and between hardware and software components. ▪ Embedded operating systems for memory management, process management, I/O, synchronization, communication.	
Q.6	Explain SOC test scheduling and test integration.	(10)
